

B. LIPIKA

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Profile Summary

Machine Learning Engineer focused on **Industrial AI, time-series forecasting, and agentic AI**. Experienced in building sensor-data analytics systems using Python, PyTorch, LangGraph, local LLMs, SQL, and backend APIs, with emphasis on evidence-driven and scalable AI workflows.

Professional Experience

AIML Trainee · *Tiny Prism Labs, Manipal*

Feb 2026 – Present

- Developing an AI-powered industrial analytics assistant that converts natural-language queries into data retrieval, analysis, and visualization workflows for multivariate sensor time-series data.
- Built a Python query-understanding pipeline that extracts equipment, metrics, operations, filters, and time ranges from natural-language requests.
- Developed Flask REST APIs and a schema-aware tool registry for SQL retrieval, dataset inspection, analytical execution, and dynamic tool invocation.
- Implemented reusable time-series analytics for trend, seasonality, stationarity, anomaly, changepoint, and correlation analysis.
- Integrated local LLMs with evidence-driven workflows, separating deterministic numerical computation from LLM-based reasoning.

Selected Projects

Industrial Time-Series Forecasting with TTM-R3

Python · PyTorch · IBM Granite TTM-R3 · Pandas · FastAPI · Time Series

- Fine-tuned IBM Granite **TTM-R3** for 15-minute industrial INFLOW forecasting using chronological train, validation, and test splits.
- Evaluated forecasting performance across long prediction horizons using **MAE, RMSE, MAPE, and sMAPE**, with repeat-period baselines for comparison.
- Performed error analysis on normal operating patterns and extreme inflow spikes to identify model strengths and forecasting limitations.
- Designed a **FastAPI forecasting service** for maintaining rolling forecasts from the latest sensor observations and serving stored short-horizon predictions through an API.

Industrial AI Agent Platform

Python · LangGraph · Local LLMs · SQL · Pandas · FastAPI

- Designed a **LangGraph-based orchestration layer** for routing industrial user queries across analytics, reasoning, and tool-execution workflows.
- Built schema-aware tools for dataset inspection, SQL querying, time-range discovery, and deterministic time-series analysis.
- Structured the system to support interchangeable LLMs and extensible tools while keeping analytical computation independent from model-generated reasoning.

AI Data Scientist Agent — Automated EDA & ML Discovery

Python · Local LLMs · AI Agents · SQL · Pandas · Scikit-learn

- Developed an AI data-science workflow for automated dataset profiling, schema discovery, analytical workflow generation, and ML experiment support.
- Implemented deterministic analytical tools for trend, seasonality, stationarity, anomaly, changepoint, correlation, and pattern analysis.
- Applied an evidence-driven architecture where numerical results are generated by Python executors and passed to the reasoning agent for interpretation.

Industrial Vibration Anomaly Detection

Python · Pandas · Parquet · Isolation Forest · Ollama

- Built an anomaly-detection workflow for a **2.3M+ row industrial vibration dataset** using statistical analysis and Isolation Forest.
- Performed data-quality analysis across vibration features and metadata, including checks for missing values, duplicates, and sensor-data consistency.
- Developed visualization workflows to examine vibration patterns and detected anomalies across industrial sensor data.
- Integrated local-LLM reasoning with analytical outputs to support interpretation of detected sensor patterns and anomalies.
- Prepared vibration features for analysis through data preprocessing and feature-level statistical examination to identify abnormal sensor behavior.

Research & Innovation

Retina AI — Visual Data to Tactile & Voice Feedback

Team Lead

Python · YOLOv8 · OpenCV · ESP32

- Developed a computer-vision assistive system combining object detection with tactile vibration and voice feedback using embedded hardware.

AgriFlow — IoT Smart Irrigation System

Indian Utility Patent No. 202541061616

ESP8266 · IoT · Sensors

- Developed a sensor-driven irrigation system with real-time soil monitoring and automated irrigation control; awarded an Indian utility patent.

Education

Bachelor of Engineering — Information Science & Engineering

2022 – 2026

Canara Engineering College, Mangalore (VTU)

CGPA: 8.25 / 10

Technical Skills

Programming	Python, SQL, Java
ML / Deep Learning	PyTorch, TensorFlow, Keras, XGBoost, LightGBM, LSTM, Scikit-learn, SHAP
Time Series	Forecasting, Trend Analysis, Seasonality, Stationarity, Anomaly Detection, Changepoint Detection, Rolling Statistics
Generative AI	LangGraph, LangChain, Ollama, Local LLMs, NLP, Tool Calling, MCP, AI Agents
Data Engineering	Pandas, NumPy, Parquet, Data Cleaning, Data Profiling, Chunk-Based Processing
Backend / Tools	FastAPI, Flask, REST APIs, Git, GitHub, Docker, Jupyter, Postman, WSL, VS Code

Certifications

- Google Data Analytics Professional Certificate — Coursera, 2025
- Java Programming — NPTEL, 2024

Achievements

- Indian Utility Patent Holder — AgriFlow Smart Irrigation System, Patent No. 202541061616
- 2nd Runner-Up — VTU Mini Project Competition